

Neuroscience Week 3: THE CEREBRUM AND HIGHER CORTICAL FUNCTION

CEREBRAL HEMISPHERE - make up the largest portion of the human brain

- highly convoluted masses of gray matter that are organized into 2 symmetrical folded structures.

SULCUS - separates gyrus

GYRUS - cortical folds

LATERAL CEREBRAL FISSURE (SYLVIAN FISSURE) separates temporal lobe from the frontal and parietal lobes.

INSULA - portion of cortex that did not grow much during development, lies deep within the fissure.

CIRCULAR SULCUS - surrounds insula and separates from diff lobes

LONGITUDINAL CEREBRAL FISSURE - separates left and right hemisphere

CENTRAL SULCUS (Fissure of Rolando) - separates frontal and parietal lobe

PARIETO-OCCIPITAL FISSURE - separate parietal from occipital lobe

CALCARINE FISSURE - begins on the medial surface of the hemisphere near occipital lobe (below splenium of corpus callosum)

CORPUS CALLOSUM - connects fissure inside

- **largest interhemispheric commissures** and is largely responsible for the activity of 2 hemispheres
- **GENU** - anterior curved portion
- **SPLENIUM** - thick posterior portion over the midbrain

FRONTAL LOBE - initiative, judgment, abstract reasoning, creativity, and socially appropriate behavior.

- extends from frontal pole to central sulcus and lateral fissure.
- **PRECENTRAL SULCUS ("before" sulcus)** - anterior to precentral gyrus; parallel to central sulcus
- **SUPERIOR AND INFERIOR FRONTAL SULCI** - 3 parallel gyri; *superior, middle, and inferior frontal gyri*
- **ORBITAL SULCI AND GYRI** - irregular
- **OLFACTORY SULCUS** - lies beneath olfactory tract
- **CINGULATE GYRUS** - crescent-shaped on the medial surface

- **PARACENTRAL LOBULE** - medial surface of hemisphere; continuation of precentral and postcentral gyri
- **PREFRONTAL CORTEX** - anterior to primary motor cortex (**BA 4,6**); **higher order association cortex**

PARIETAL LOBE - somatosensory sensory relay

- **POSTCENTRAL SULCUS** - behind postcentral gyrus
- **INTRAPARIETAL SULCUS** - horizontal groove that sometimes unite with postcentral sulcus
- **SUPERIOR PARIETAL LOBULE** - lies above horizontal portion of intraparietal sulcus
- **SUPRAMARGINAL GYRUS** - arches above ascending end of posterior ramus of lateral cerebral fissure
- **ANGULAR GYRUS** - above end of superior temporal sulcus and is continuous and with middle temporal gyrus

OCCIPITAL LOBE (BA 17,18,19) - houses **primary visual cortex**; behind parieto-occipital fissure

- **CALCARINE FISSURE** - divides medial surface of occipital lobe into **cuneus and lingual gyrus**

TEMPORAL LOBE - hearing, balance, initiation of language; below lateral cerebral fissure

- **LATERAL SURFACE** - superior, middle, inferior
- **TRANSVERSE TEMPORAL GYRUS** - posterior part of superior temporal surface
- **HIPOCCAMPAL GYRUS** - inferomedian aspect of lobe from area of splenium of corpus callosum to uncus; for emotion
- **UNCUS** - most medial portion of temporal lobe; curves in form of hook

INSULA - sunken portion of cerebral cortex (looks like an underdeveloped 5th lobe)

LIMBIC SYSTEM - responsible for emotion

- CINGULATE
- PARAHIPPOCAMPUS
- SUBCALLOSAL GYRI
- HIPPOCAMPAL FORMATION

WHITE MATTER OF CEREBRUM - contains **myelinated fibers** (mostly **oligodendrocytes**)

- sometimes called **CENTRUM SEMIOVALE**
- contains myelinated transverse fibers, projection fibers, and association fibers
- **TRANSVERSE (COMMISSURAL FIBER)** - interconnect 2 cerebral hemisphere
 - **ANTERIOR COMMISSURE** - connects to olfactory bulbs and temporal lobe structures
 - **HIPPOCAMPAL FORMATION** - joins two hippocampi

- **PROJECTION FIBERS** - connect cerebral cortex with lower portions of brain/spinal cord
- **ASSOCIATION FIBERS** - connect various portions of cerebral hemisphere and permit cortex to function as a coordinated whole

3 MAIN TYPES OF NEURONS

1. PYRAMIDAL CELLS
2. STELLATE NEURONS
3. FUSIFORM NEURONS

TYPES OF CORTEX IN CEREBRAL CORTEX

1. **ALLOCORTEX** - in limbic system and contains fewer layers than isocortex
2. **ISOCORTEX (NEOCORTEX)** - in cerebral hemisphere and contains 6 layers
3. **JUXTALLOCORTEX (MESOCORTEX)** - transition between allocortex and isocortex

LAYERS OF ISOCORTEX

1. Molecular Layer
2. External Granular Layer
3. External Pyramidal Layer
4. Internal Granular Layer
5. Internal Pyramidal Layer
6. Fusiform (Multiform) Layer

BRODMANN'S CLASSIFICATION - common in naming the functional areas of cerebrum

- reference base for the localization of physiologic and pathologic processes
- **FRONTAL LOBE**
 - **AREA 4 (Primary Motor Area)** - voluntary muscle activation, contributes to **Corticospinal Tract**, represented by motor homunculus (right)
 - **AREA 6 (Premotor Area)** - planning and organizing movements and actions
 - **AREA 8 (Frontal Eye Field)** - eye movement
 - **AREA 44 & 45 (Broca's Area)** - motor aspects of speech production, initiation, and articulation
- **PARIETAL LOBE**
 - **AREA 3,1,2 (Primary Sensory Area)** - process sensory from external environment
- **OCCIPITAL LOBE**
 - **AREA 17,18,19 (Primary Visual Cortex)** - processing of visual stimuli
- **TEMPORAL LOBE**
 - **AREA 41 (Primary Auditory Cortex)**
 - **AREA 42 (Associative Auditory Cortex)**

- Note: both are involved in processing auditory stimuli; referred to as **Heschl's gyrus**
- **AREA 22 - Language Comprehension**
- **PLANUM TEMPORALE** - located on superior surface of temporal lobe; involved in language and music

PRIMARY MOTOR CORTEX - corresponding generally to the distribution of the **giant pyramidal (Betz's) cells** (main cells of corticospinal tract)

- **GIANT PYRAMIDAL (BETZ'S) CELLS** - involved in corticobulbar (head and neck) and corticospinal tract
- **Clinical Correlation**
 - **IRRITATIVE LESIONS** - can cause lesions that begins as **focal twitching (somatotopic manner)** to involve larger muscle groups (**Jacksonian epilepsy**)
 - **DESTRUCTIVE LESIONS** - contralateral **flaccid (relaxed)** paresis or paralysis of affected muscle groups
 - **AREA 6** - **spastic (tight)** paralysis

PRIMARY SENSORY CORTEX - sensory information received from the skin, mucosa, and other tissues of the body and face

- Located in postcentral gyrus and is called the **somesthetic area (Area 3,1,2)**
- **Convey touch and proprioceptive** (muscle, joint, and tendon) sensations from the **Thalamus** (from opposite side of body)
 - the only information that doesn't go through Thalamus: **Olfactory (CN I)**
- Wide portion of adjacent primary and frontal and parietal lobes.
- Considered as **secondary sensory cortex**
- **PRIMARY SENSORIMOTOR AREA** - functioning as both a motor and a sensory cortex.
- **CORTICAL TASTE AREA** - receives gustatory information relayed from solitary nucleus in the medulla via ventral posteromedial nucleus and
- **Clinical Correlation**
 - **IRRITATIVE LESION** - produce **PARESTHESIA** on the opposite side of the body
 - **DESTRUCTIVE LESIONS** - subjective and objective impairments in sensibility

PRIMARY VISUAL CORTEX AND VISUAL ASSOCIATION CORTEX - in occipital lobe

- **PRIMARY VISUAL CORTEX AREA 17**
 - **POSTERIOR PORTION** - **high-resolution macular** vision
 - **ANTERIOR PORTION** - **peripheral** vision

- **VISUAL ASSOCIATION AREA - AREA 18 & 19**
- **Clinical Correlation**
 - **IRRITATIVE LESIONS** - hallucinations (flashes of light, rainbows, brilliant starts).
 - **DESTRUCTIVE LESIONS** - contralateral homonymous defects of the visual fields without macular involvement (since it's present in retina).
 - **INJURY TO AREA 18,19**

PRIMARY AUDITORY RECEPTIVE CORTEX - in transverse temporal gyrus

- Receives **auditory radiation from cochlea**
- **LOW TONES** - high tones (Area 41)
- **AREA 22 (WERNICKE'S AREA)** - high ordered auditory discrimination and speech comprehension
- **Clinical correlation:**
 - buzzing and roaring sensation
 - **mild hearing loss (*unilateral lesion*)**
 - **deafness (*bilateral lesion*)**
 - area 22 - pure word deafness—**Wernicke's Aphasia**

BASAL GANGLIA - essential function role in motor control.

- **CAUDATE NUCLEUS** - elongated gray mass whose pearshaped head is continuous and with putamen.
- **PUTAMEN**
- **GLOBUS PALLIDUS**
- **INTERNAL CAPSULE** - imparting a striped appearance.
 - separates lentiform nucleus from medial caudate nucleus and thalamus.
 - consists of **anterior and posterior limb**
 - **ANTERIOR LIMB** - separates **lentiform nucleus from caudate nucleus**; more on **sensory** parts
 - **POSTERIOR LIMB** - contains major ascending and descending pathways; **lacunar infarct; pure motor strokes**
 - contains pathways such as **corticobulbar and corticospinal (down to spine) tract**
 - **CORTICOSPINAL TRACTS** - lower motor neuron
 - voluntary, highly skilled and fractioned movement.
 - **CORTICOBULBAR TRACTS** - controlling voluntary movements of the **muscles of head and neck (facial expressions, chewing, swallowing, and speech).**

- **CORPUS STRIATUM** - termed given to **caudate nucleus and putamen**
- **STRIATUM** - when **caudate nucleus and putamen develop together** and contain similar cells
- **LENTICULAR NUCLEI** - lens shaped mass formed **from putamen and globus pallidus**
- **LENTICULAR NUCLEUS** - between insula and internal capsule; **divided by external medullary lamina**
- **CLAUSTRUM AND EXTERNAL CAPSULE** - thin layer of gray substance beneath insular cortex

HIGHER CORTICAL FUNCTIONS

- **FRONTAL LOBE FUNCTIONS**
 - Damage to Frontal Lobes
 - **DORSOLATERAL** - indifferent, abulic or apathetic.
 - **ORBITOFRONTAL AREA** - loss of usual inhibition and social graces.
 - **MEDIAL - AKINESIA**: lack of spontaneous movement and apathy.
 - **BASAL** - memory impairment
- **LANGUAGE AND SPEECH**
 - **LANGUAGE** - comprehension and communication of abstract ideas.
 - **BROCA'S AREA**
 - **WERNICKE'S AREA** - ability to comprehend language.
 - **ARCUATE FASCICULUS** - connects Broca's and Wernicke's area; damage to this matter produces impairment of repetition.
 - **DYSARTHRIA** - speech disorder characterized by dysfunction of phonation, articulation, resonance or respiration aspects of speech (*utal-utal*).
 - **APHASIA** - language function impairment.
 - **CATEGORIES OF SPEECH:**
 - **FLUENT** - more than 50 words per minute, effortless,
 - **NON-FLUENT** - effortful with less than 50 words.
 - **BROCA'S APHASIA** - affects middle cerebral artery territory
 - **WERNICKE'S APHASIA** - has difficulty in naming objects.
 - **Neologisms** - meaningless, nonsensical words
 - **GLOBAL APHASIA** - repetition and comprehension are severely impaired.
 - **CONDUCTION APHASIA** - comprehension is intact but repetition and naming are impaired.

- **ISOLATION APHASIA** - repetition is spared but comprehension is impaired; referred to as **TRANSCORTICAL APHASIAS**
- **ANOMIC APHASIA**
 - **ANOMIA** - difficulty in finding the right word.
- **ALEXIA** - inability to read
- **ALEXIA WITH AGRAPHIA** - difficulty in reading and writing.
- **AGNOSIA** - difficulty in identification or recognition.
 - **VISUAL AGNOSIA** - inability to recognize things by sight.
 - **ASTEREOGNOSIS** - failure of tactile recognition (ability to recognize objects without looking at it) of objects.
 - **PROSOPAGNOSIA** - inability to recognize familiar faces.
 - **UNILATERAL NEGLECT** - fails to respond to stimuli in one half of space, contralateral to a hemispheric lesion (ex: when tying a shoe, a patient doesn't tie the left/right shoe); usually in the right cerebral hemisphere.
- **ANOSOGNOSIA** - denial of illness.
- **APRAXIA** - inability to carry out motor acts correctly despite intact motor and sensory pathways.
 - **IDEOMOTOR APRAXIA** - inability to perform motor responses upon verbal command.
 - **IDEATIONAL APRAXIA** - difficulty doing anything at all.
- **GERTSMAN SYNDROME** - dysfunction in the angular gyrus of the left hemisphere.

- **MEMORY AND LEARNING**

- **IMMEDIATE** - allows people to repeat a small amount of information shortly after hearing/reading it; lasts up to 10mins.
- **SHORT-TERM MEMORY** - last up to an hour
- **LONG-TERM MEMORY** - allows people to remember words, numbers, etc for many years.
 - **LONG TERM POTENTIATION (LTP)**
 - **NOTE:** Immediate and short-term memory - neurons that die when a person is dying. So they only see long-term ones.
- **EPILEPSY** - alterations in consciousness; generalized/partial.

- **FOCAL (JACKSONIAN) EPILEPSY** - focal motor seizures (palaki nang palaki).
 - **COMPLEX PARTIAL EPILEPSY** - automatisms, simple, or patterned movements, writhing movements. (when a person experiences a seizure, he doesn't remember it afterwards).
 - **TEMPORAL LOBE EPILEPSY** - abnormal feeling of familiarity "*deja vu*"